

Expert-Inspired Multi-Agent Coordination for Multi-Objective Molecular Optimization

Daojian Zeng¹, Tianle Li¹, Jiahao Yang^{2,3}, Jiakai Yi⁴, Xieping Gao¹, Lincheng Jiang^{5,*}, Tengfei Ma^{2,*}, Xiangxiang Zeng²

¹Hunan Provincial Key Laboratory of Philosophy and Social Sciences of Artificial Intelligence and International Communication, Hunan Normal University Changsha 410081, China

²State Key Laboratory of Chemo and Biosensing, College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China

³Lingang Laboratory, Shanghai 201306, China

⁴School of Computer Science, National University of Defense Technology, Changsha 410073, China

⁵College of Advanced Interdisciplinary Studies, National University of Defense Technology, Changsha 410073, China
{zengdj, linjiadegou2}@hunnu.edu.cn, tfma@hnu.edu.cn, linc Chengjiang@nudt.edu.cn

Abstract

Multi-objective molecular optimization is a fundamental yet inherently challenging task in drug discovery, as it requires simultaneously optimizing multiple, often conflicting, molecular properties. Although recent deep learning methods have shown promise, they often lack objective-specific specialization and dynamic coordination, which makes them ineffective at handling competing objectives and limits their scalability in complex, high-dimensional molecular design tasks. Inspired by the division of labor among domain experts in medicinal chemistry, we propose **MAMO**, a multi-agent framework for molecular design that simulates expert collaboration. Each agent specializes in optimizing a single objective, and their interactions are orchestrated by a central scheduling module that dynamically reallocates tasks based on evaluation feedback. This coordination mechanism enables interpretable and goal-conditioned optimization while adaptively balancing conflicting objectives. Extensive experiments on benchmark datasets demonstrate that MAMO consistently achieves superior performance in both objective quality and Pareto diversity, particularly in scenarios with strong inter-objective conflict. Our results highlight the potential of multi-agent coordination strategies for scalable and conflict-aware molecular design.

Code — <https://github.com/linjiadegou/MAMO>

Introduction

Multi-objective molecular optimization plays a pivotal role in early-stage drug discovery (De Rycker et al. 2018), where the goal is to refine a given lead compound to simultaneously satisfy multiple desirable properties such as bioactivity, drug-likeness, and synthesizability (He et al. 2021). Traditional approaches, including high-throughput screening (HTS) (Graff, Shakhnovich, and Coley 2021) and simulation-based techniques (Hsu et al. 2017), are inherently time-consuming and resource-intensive.

*Corresponding authors.

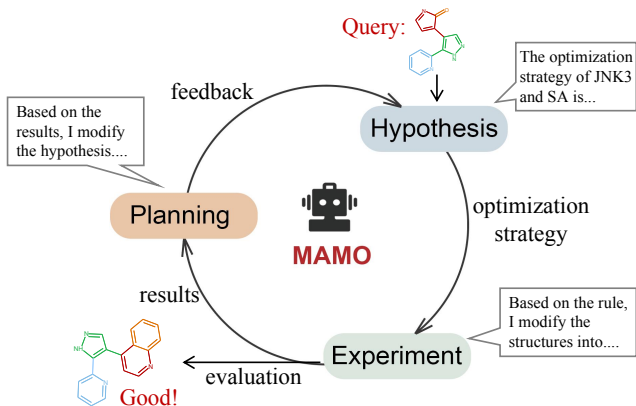


Figure 1: Conceptual illustration of MAMO. Inspired by expert-driven scientific workflows, MAMO follows a closed-loop process of hypothesis generation, structural modification, evaluation, and planning to iteratively optimize molecules across multiple conflicting objectives.

Influenced by the rapid development of artificial intelligence, machine learning has demonstrated strong potential in efficient candidate discovery (Hoffman et al. 2022). Previous methods for multi-objective molecular design typically reduce the problem to a single-objective formulation by assigning predefined weights to individual objectives (Ji et al. 2021; Xia et al. 2024). While effective in some settings, these methods heavily depend on expert-designed weightings, which are often difficult to calibrate and may lead to suboptimal solutions when objectives conflict (Xie et al. 2021; Fromer and Coley 2023). Alternatively, some approaches aim to identify the Pareto front through a two-stage process involving large-scale molecular sampling and non-dominated sorting (Verhellen 2022). However, such methods are computationally expensive and scale poorly with the number of objectives and candidate molecules. To address efficiency concerns, Bayesian optimization has been widely adopted due to its sample efficiency and principled

uncertainty modeling (Xie et al. 2021; Gao et al. 2022), yet it struggles with high-dimensional spaces and costly Gaussian process inference, limiting its practical use in multi-objective molecular design.

With the rapid advances in large language models (LLMs) (OpenAI et al. 2024; Bai et al. 2023; Dubey et al. 2024), recent research has begun to explore their potential in molecular generation (Brahmavar et al. 2024; Wang, Yang, and Shen 2025) and optimization (Yu et al. 2025; Ye et al. 2025). LLMs offer a flexible and scalable alternative, with the capacity to perform goal-conditioned generation and reasoning across diverse property objectives (Nguyen and Grover 2024). However, existing LLM-based methods often treat multi-objective optimization as a monolithic task, lacking explicit mechanisms for balancing trade-offs among conflicting goals (Liu et al. 2025).

To address the limitations of monolithic LLM-based approaches in managing trade-offs across multiple molecular objectives, we take inspiration from the collaborative workflows in medicinal chemistry (Pitt et al. 2025). In real-world lead optimization, domain experts such as biologists, pharmacokineticists, and synthetic chemists each contribute by focusing on distinct property dimensions, including activity, ADME, and synthetic accessibility. These experts iteratively refine molecular candidates through cycles of hypothesis generation, experimental testing, and adaptive planning (Kapustina et al. 2024; Mathur et al. 2025). Building on this paradigm, we present **MAMO**, a multi-agent framework that simulates expert collaboration through a closed-loop optimization process (Figure 1). To operationalize this process, MAMO is organized into four coordinated modules: **Formulation**, **Collaboration**, **Evaluation**, and **Scheduling**, each of which plays a distinct role in enabling iterative and goal-driven molecular optimization (see Section Method). At each iteration, LLM-based expert agents generate hypotheses and propose property-specific structural modifications based on their assigned optimization goals. These candidates are then evaluated by external tools, and results are fed back to a central scheduling agent that adaptively re-allocates priorities for the next iteration. This iterative process mirrors the scientific discovery loop and allows MAMO to flexibly balance conflicting objectives, support modular reasoning, and converge on high-quality molecules. By integrating LLMs with structured agent coordination, MAMO offers an effective solution to multi-objective molecular optimization. In summary, our contributions include:

- We propose **MAMO**, an expert-inspired multi-agent framework that simulates domain-specific collaboration for multi-objective molecular optimization, addressing the intrinsic conflict between properties.
- We introduce a **centralized scheduling agent** that dynamically allocates optimization tasks based on real-time evaluation feedback, enabling adaptive coordination and effective trade-off balancing among expert agents.
- Extensive experiments on benchmark tasks demonstrate that MAMO consistently outperforms strong baselines in both optimization quality and objective balancing, particularly under high-conflict, high-dimensional scenarios.

Related Work

Conventional molecular optimization. Molecular optimization is a fundamental task in drug discovery and has evolved from manual experiments to computational methods (Gao et al. 2022). Current approaches fall into two main categories: (1) *Generative models*, including variational autoencoders (Gómez-Bombarelli et al. 2018; Jin, Barzilay, and Jaakkola 2018), GANs (Guimaraes et al. 2017; Kadurin et al. 2017), and diffusion models (Ho, Jain, and Abbeel 2020; Xu et al. 2022), which generate candidate molecules in a data-driven manner; and (2) *Combinatorial optimization*, which searches chemical space using techniques like Monte Carlo tree search (Yang et al. 2017), genetic algorithms (Fu et al. 2022), and reinforcement learning (Olivecrona et al. 2017; Gu and Dao 2023). Despite their progress in single-objective tasks, these methods often lack the flexibility needed to balance conflicting objectives.

LLM for molecular optimization. Recent advances in large language models (LLMs) have opened up new opportunities for molecule-related tasks, including property prediction and molecular generation (Luo et al. 2022; Yunxiang et al. 2023; Han et al. 2023; Wu et al. 2024; Fang et al. 2023). Emerging efforts have also explored the application of LLMs in molecular optimization. For instance, MOLLEO (Wang et al. 2024) integrated LLMs into genetic algorithms by using them as operators for crossover and mutation, improving both solution quality and convergence speed. LICO (Nguyen and Grover 2024) employed context-aware prompting for molecule refinement, enabling in-context optimization without retraining. DrugAssist (Ye et al. 2023) further proposed an interactive human-in-the-loop optimization framework built on LLMs to leverage human knowledge and LLM reasoning collaboratively. Despite these initial efforts, most existing LLM-based methods treat multi-objective optimization as a unified generation task, lacking explicit modeling of inter-objective conflicts and coordination. The full potential of LLMs for adaptive and multi-agent optimization in complex design tasks remains largely untapped.

Multi-agent LLM systems. Large language models (LLMs) have become core components of autonomous agents for complex reasoning and decision-making tasks (Chen et al. 2023). Early systems such as AutoGPT (Yang, Yue, and He 2023) demonstrated tool-augmented task execution but were limited to single-agent settings, making them less effective for solving multi-dimensional or interdependent tasks. To overcome these limitations, recent work has explored multi-agent collaboration with LLMs. AutoGen (Wu et al. 2023) enables structured communication among role-specific agents, while MetaGPT (Hong et al. 2023) improves task planning and execution through coordinated agent roles. These approaches demonstrate the potential of multi-agent LLM frameworks for solving complex tasks. In particular, tasks such as multi-objective molecular optimization can benefit from expert-like role decomposition and collaboration, which remain underexplored in current systems.

Method

Problem Definition. We formulate multi-objective molecular optimization as a discrete optimization problem over the chemical space $\mathcal{X}$, where each molecule is represented by a valid SMILES string. Given an initial molecule $x_0 \in \mathcal{X}$, the goal is to generate a set of molecules $\{x_1, x_2, \dots, x_n\} \subset \mathcal{X}$ that jointly optimize multiple properties. Formally, we aim to solve a K -objective maximization problem:

$$\max_{x \in \mathcal{X}} \mathbf{f}(x) = [f_1(x), f_2(x), \dots, f_K(x)], \quad (1)$$

where each $f_k(x) : \mathcal{X} \rightarrow \mathbb{R}$ denotes a non-differentiable scoring function for a molecular property, such as GSK3 β /JNK3 inhibition, drug-likeness (QED), and synthetic accessibility (SA). The objective is to approximate the Pareto front $\mathcal{P} \subset \mathcal{X}$, i.e., the set of non-dominated molecules. We enforce chemical validity as a hard constraint to ensure all generated molecules are syntactically and chemically feasible.

Overview of MAMO. We introduce **MAMO**, a modular multi-agent framework for multi-objective molecular optimization. As shown in Figure 2a, MAMO comprises four key components: *Formulation*, *Collaboration*, *Evaluation*, and *Scheduling*. Given a molecule in SMILES format (Weininger 1988), each expert agent—powered by a large language model (LLM)—focuses on optimizing a specific molecular property (e.g., QED, SA). Agents collaboratively propose candidates based on property-aware strategies, which are evaluated by external tools. A scheduling module adaptively coordinates agent interactions based on feedback to guide the search process. This design enables goal-directed, scalable optimization across conflicting objectives. The functional roles of agents, grounded in domain knowledge, are illustrated in Figure 2b.

Formulation. *Agent Initialization:* As illustrated in Figure 2a, we instantiate a set of expert agents, each dedicated to optimizing a specific molecular property. Each agent is built upon an LLM, such as GPT-4o mini (OpenAI et al. 2024), and is assigned a domain-specific role, including QED Expert, SA Expert, GSK3 β /JNK3 (targets related to Alzheimer’s) Experts, according to the optimization target (Figure 2b). Recent studies demonstrate that the performance of LLMs on biochemical tasks can be significantly improved through domain-aware prompt engineering (Li et al. 2025; Luo et al. 2025). Following this insight, we design tailored prompts for each agent to explicitly define the task scope, optimization objective, input format, and expected output structure. An example prompt for the QED expert is shown in Figure 3. To further enhance agent specialization and actionability, we equip each expert with tool-calling capabilities, allowing access to domain tools such as RDKit and property oracles for informed decision-making. Details on tool usage are provided in Appendix A. *Optimization Strategy Formulation:* MAMO adopts a progressive optimization mechanism that decomposes molecular design into an iterative sequence of strategy-driven steps. At each stage, expert agents adapt their behavior based on feedback from prior rounds, forming a coherent and adaptive

optimization trajectory. This design allows the framework to maintain logical consistency while flexibly adjusting optimization plans to match the molecular context and evolving property profiles. For each objective, the corresponding expert agent generates a strategy through prompt-based reasoning. These strategies are dynamically tailored to the characteristics of the input molecule, ensuring that modifications are both targeted and interpretable. By iteratively refining the molecule toward predefined property goals, the system effectively narrows the gap between current and desired performance. Additionally, strategy prompts are updated at each step based on multi-agent feedback, enabling agents to respond adaptively to emerging trade-offs (Appendix B).

Collaboration. To enable effective coordination among expert agents during molecular optimization, MAMO introduces a shared-memory mechanism where all agents access and update a common molecular context. This shared memory stores intermediate molecules, strategy updates, and property evaluations, allowing agents to (i) avoid redundant edits, (ii) recognize emerging trade-offs, and (iii) adapt their context-driven strategies based on global optimization dynamics. However, as the number of objectives increases, simple peer-to-peer interaction becomes inefficient. To address this, MAMO incorporates a centralized scheduling agent that monitors the shared memory and dynamically assigns optimization tasks to experts based on real-time progress and remaining objectives. This scheduler improves coordination by minimizing inter-agent conflicts, accelerating convergence, and enhancing the quality of the Pareto front. The scheduling mechanism is detailed in Section Central Scheduling.

Evaluation. Following each round of multi-agent collaboration, MAMO employs a dedicated *evaluation agent* to assess the quality of generated candidate molecules across all relevant property dimensions. This agent operates independently of the expert agents and is responsible for verifying whether the current candidates meet the predefined optimization goals. To perform this assessment, the evaluation agent dynamically selects appropriate property prediction tools—such as RDKit-based QED or synthetic accessibility calculators—based on the task configuration. It focuses only on the properties under active optimization and computes the corresponding scores to determine task satisfaction. If any objective remains unmet, the evaluation agent generates structured feedback and forwards it to the central scheduling agent. This feedback is then used to adjust agent priorities and revise optimization strategies in subsequent rounds. The evaluation process ensures closed-loop coordination between local optimization and global objective satisfaction. Detailed tool configurations and the prompt design of the evaluation agent are provided in Appendix C.

Central Scheduling. In the multi-agent collaboration framework, we define the set of candidate molecules as

$$\mathcal{M} = \{x_1, x_2, \dots, x_N\}, \quad (2)$$

and the set of expert agents as

$$\mathcal{E} = \{e_1, e_2, \dots, e_K\}, \quad (3)$$

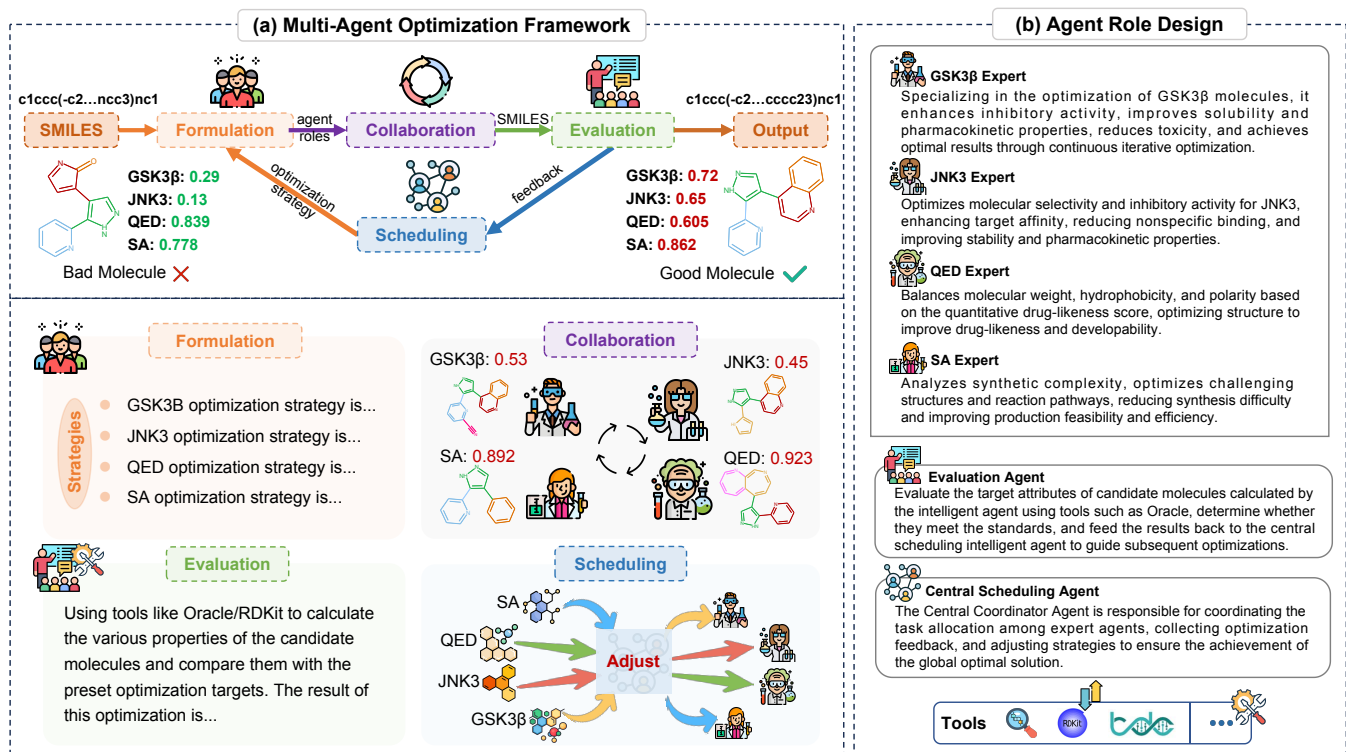


Figure 2: Overview of the **MAMO** framework. (a) A multi-agent optimization pipeline where each expert agent focuses on a specific molecular property. The system iteratively refines candidates through formulation, collaboration, evaluation, and dynamic scheduling. (b) Role definitions and tool configurations for expert, evaluation, and scheduling agents.

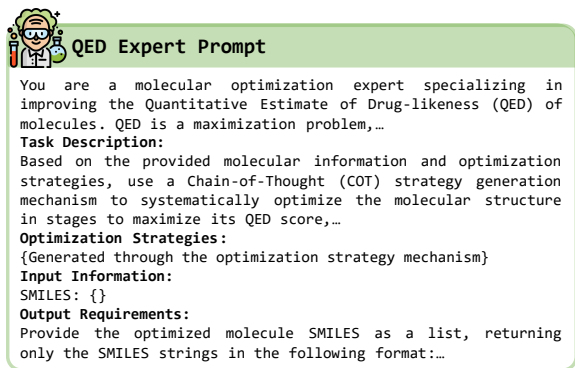


Figure 3: Prompt Template for the QED Expert Agent.

where N is the number of candidate molecules and K indicates the number of optimized objectives. The optimization proceeds in discrete rounds $t = 1, 2, \dots, T$. After each round, each expert agent e_j independently optimizes the assigned molecules. A central scheduling agent (referred to as the scheduler) is responsible for task scheduling and allocation to enhance the efficiency of global optimization. After each round t , an evaluation agent provides feedback on molecular properties, represented as a property vector:

$$\mathbf{p}_i^t \in \mathbb{R}^K, \quad (4)$$

where $\mathbf{p}_i^t$ is the property values of i -th molecule from t -th round. Given a target property vector $\mathbf{p}^* \in \mathbb{R}^K$, we define the optimization gap as

$$g_i^t = d(\mathbf{p}_i^t, \mathbf{p}^*), \quad (5)$$

where $d(\cdot, \cdot)$ is defined as the Euclidean distance. The normalized priority score for each molecule is calculated as

$$\pi_i^t = \frac{g_i^t}{\sum_{j=1}^N g_j^t}. \quad (6)$$

Molecules with larger π_i^t are considered more critical for optimization in the next round. To allocate tasks for round $t+1$, the scheduler dynamically assigns molecules to agents based on the priority scores π_i^t . This ensures that each agent is assigned exactly one molecule, and each molecule is assigned to at most one agent, maximizing overall optimization priorities. The scheduler does not directly engage in molecular optimization but dynamically adjusts allocations based on updated evaluations $\{\mathbf{p}_i^t\}$ and molecule dependencies. Throughout the optimization process, the system iteratively executes the sequence:

$$\{A^1, A^2, \dots, A^T\} \xrightarrow{\text{scheduling}} \text{global convergence} \quad (7)$$

By separating scheduling from optimization, the proposed multi-agent framework achieves higher flexibility and scalability in complex molecular design tasks. Detailed prompt designs for the scheduler are provided in Appendix D.

Methods	QED+SA	GSK3 β +JNK3	JNK3+QED+SA	GSK3 β +QED+SA	GSK3 β +JNK3+QED	GSK3 β +JNK3+QED+SA	Sum ($\uparrow$)
REINVENT	0.894 $\pm$ 0.000	0.145 $\pm$ 0.000	0.476 $\pm$ 0.005	0.335 $\pm$ 0.000	0.057 $\pm$ 0.006	0.092 $\pm$ 0.000	1.999
Graph GA	0.934$\pm$0.002	0.302 $\pm$ 0.067	0.240 $\pm$ 0.073	0.526 $\pm$ 0.045	0.185 $\pm$ 0.076	0.342 $\pm$ 0.096	2.529
SMILES GA	0.876 $\pm$ 0.003	0.097 $\pm$ 0.011	0.093 $\pm$ 0.004	0.377 $\pm$ 0.077	0.085 $\pm$ 0.016	0.070 $\pm$ 0.021	1.598
GP BO	0.925 $\pm$ 0.001	0.241 $\pm$ 0.043	0.201 $\pm$ 0.039	0.466 $\pm$ 0.030	0.320 $\pm$ 0.110	0.304 $\pm$ 0.101	2.457
REINVENT Trans	0.931 $\pm$ 0.002	0.478 $\pm$ 0.058	0.432 $\pm$ 0.059	0.628 $\pm$ 0.068	0.358 $\pm$ 0.088	0.270 $\pm$ 0.057	3.097
DrugAssist	0.812 $\pm$ 0.002	0.583 $\pm$ 0.012	0.565 $\pm$ 0.036	0.597 $\pm$ 0.045	0.387 $\pm$ 0.023	0.444 $\pm$ 0.063	3.388
EAG	0.912 $\pm$ 0.001	0.669 $\pm$ 0.021	0.665 $\pm$ 0.043	0.631 $\pm$ 0.061	0.585 $\pm$ 0.069	0.517 $\pm$ 0.075	3.979
GPT-4o mini	0.851 $\pm$ 0.003	0.612 $\pm$ 0.021	0.605 $\pm$ 0.055	0.574 $\pm$ 0.072	0.564 $\pm$ 0.101	0.501 $\pm$ 0.074	3.707
MAMO	0.928 $\pm$ 0.001	0.728$\pm$0.018	0.732$\pm$0.067	0.728$\pm$0.041	0.718$\pm$0.062	0.676$\pm$0.031	4.510

Table 1: Performance (HV) comparison across methods. The **bold** and underline denote the best and second-best scores.

	QED	SA	GSK3 β	JNK3
QED	1	0.051	-0.094	-0.220
SA	0.051	1	0.064	-0.055
GSK3 β	-0.094	0.064	1	0.351
JNK3	-0.220	-0.055	0.351	1

Table 2: The correlation between multiple objectives.

EXPERIMENTS

Experimental Setup

Implementation Details. We consider four molecular property objectives and analyze their pairwise correlations on a subset of the ZINC20 dataset (Irwin et al. 2020). As shown in Table 2, most objective pairs exhibit low correlation, indicating substantial inter-objective independence. Based on this, we design five task settings with varying conflict intensities: 1) QED + SA (non-biological objectives): Drug-likeness (QED) and synthetic accessibility (SA) measure the developability and synthesizability of molecules, computed using RDKit. 2) GSK3 β + JNK3 (biological objectives): The inhibition of GSK3 β and JNK3, two kinase targets associated with Alzheimer’s disease, predicted using random forest models. 3) QED + SA + GSK3 β /JNK3: Optimization of either GSK3 β or JNK3 inhibition under constraints of good QED and SA properties. 4) GSK3 β + JNK3 + QED: Simultaneous optimization of GSK3 β and JNK3 inhibition with QED constraints, without explicit control over synthetic accessibility. 5) QED + SA + GSK3 β + JNK3: Joint optimization across all four objectives to balance activity, drug-likeness, and synthesizability.

Dataset and Metrics. We adopt the RationaleRL dataset (Jin, Barzilay, and Jaakkola 2020) to ensure a diverse initial molecular population for multi-objective optimization across GSK3 β inhibition, JNK3 inhibition, QED, and SA. To assess optimization performance, we primarily use the hypervolume (HV) metric (Zitzler et al. 2003), which measures the volume of the dominated region in objective space relative to a reference point. HV effectively captures both convergence toward the Pareto front and the diversity of the solution set. In addition, we report standard evaluation metrics in molecular generation, including novelty and diversity. (details in Appendix F).

Baselines. We compare **MAMO** against both traditional baselines and LLM-based frameworks. The traditional

methods include: 1) **REINVENT** (Olivecrona et al. 2017), which employs reinforcement learning to guide an RNN-based generative model; 2) **Graph GA** (Jensen 2019), a genetic algorithm operating on molecular graph structures; 3) **SMILES GA** (Brown et al. 2019), which performs genetic optimization over SMILES representations; 4) **GP BO** (Tripp, Simm, and Hernández-Lobato 2021), a Gaussian process-based Bayesian optimization method; and 5) **REINVENT Trans** (Pre-training and Fine-tuning 2024), a transformer-based variant of REINVENT. The LLM-based baselines include: 1) **DrugAssist** (Ye et al. 2025), which formulates molecule optimization as an interactive dialogue with an LLM; 2) **EAG** (Gu et al. 2025) decomposes complex tasks into pipeline stages to enhance multi-agent coordination; and 3) a direct **GPT-4o mini**-based optimization baseline, which applies a single-agent LLM to optimize multiple objectives simultaneously (details in Appendix G).

Main Results

Performance Comparison on Multi-Objective Tasks. Table 1 reports HV scores across six multi-objective molecular optimization tasks. MAMO consistently achieves the best performance in most settings, yielding the highest HV scores in each objective combination and the largest overall HV sum (4.51). This demonstrates its superior ability to explore diverse high-quality solutions and effectively balance conflicting objectives. On simpler dual-objective tasks such as QED+SA and GSK3 β +JNK3, most baselines perform reasonably well, especially Graph GA and REINVENT Trans. However, their performance declines as objectives increase. Notably, SMILES GA and REINVENT perform poorly on tasks involving biological targets, highlighting their limited scalability in high-dimensional spaces. Among LLM-based approaches, single-agent methods like DrugAssist (HV sum: 3.39) and direct GPT-4o mini optimization (HV sum: 3.71) show constrained capability in handling objective conflicts. While the multi-agent framework EAG (HV sum: 3.98) demonstrates improved coordination, MAMO’s specialized expert collaboration and dynamic scheduling significantly outperform all these LLM-based alternatives.

Performance on Top-ranked Candidate Molecules. To further understand the effectiveness of each method on individual molecular properties, we evaluate the top-10 and top-50 average scores for each objective, as shown in Tables 3 and 4. We also report the overall average score and average rank for each method. In the biological objective

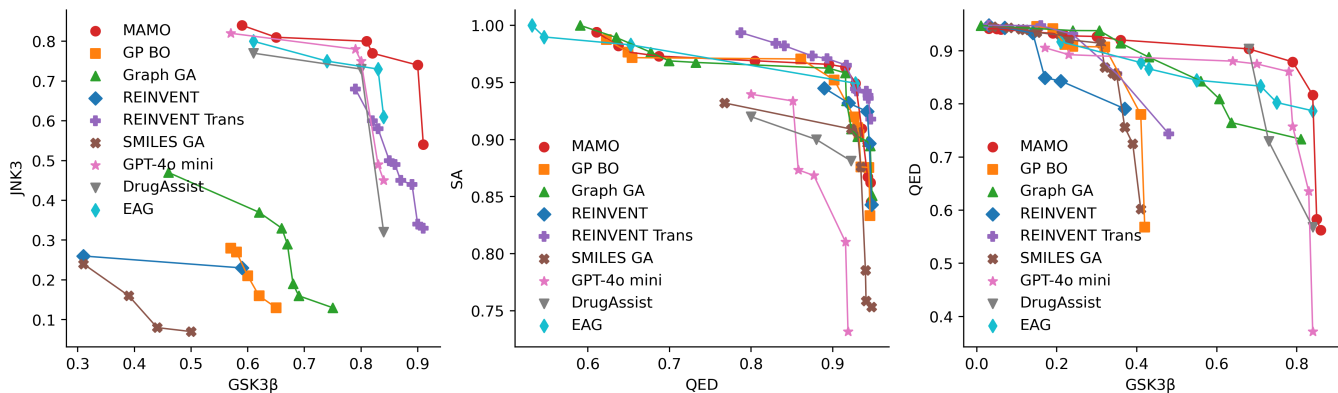


Figure 4: Non-dominated solutions of various methods on $\text{GSK3}\beta$ +JNK3, QED+SA, and $\text{GSK3}\beta$ +QED objectives.

Methods	Top-10 Avg $\text{GSK3}\beta$	Top-10 Avg JNK3	Top-50 Avg $\text{GSK3}\beta$	Top-50 Avg JNK3	Avg $\text{GSK3}\beta$	Avg JNK3	Avg Rank
REINVENT	0.495	0.234	0.395	0.167	0.077	0.036	9
Graph GA	0.688	0.343	0.636	0.244	0.296	0.061	7
SMILES GA	0.448	0.211	0.383	0.181	0.164	0.067	8
GP BO	0.619	0.229	0.586	0.179	0.331	0.059	6
REINVENT Trans	0.880	0.616	0.865	0.557	<u>0.429</u>	0.217	5
DrugAssist	0.791	0.698	0.729	0.593	0.382	0.255	4
EAG	0.812	<u>0.723</u>	0.755	<u>0.635</u>	0.412	<u>0.282</u>	2
GPT-4o mini	0.801	0.708	0.739	0.633	0.408	0.268	3
MAMO	<u>0.846</u>	0.798	<u>0.810</u>	0.740	0.442	0.322	1

Table 3: Performance of MAMO on top-ranked molecules under $\text{GSK3}\beta$ +JNK3.

pair ($\text{GSK3}\beta$ + JNK3), MAMO achieves the highest average rank (1.0), with top-10 and top-50 scores of 0.846 and 0.810 for $\text{GSK3}\beta$, and 0.798 and 0.740 for JNK3, respectively. While REINVENT Trans performs strongly on $\text{GSK3}\beta$ (Top-50: 0.865), its performance on JNK3 is significantly lower (Avg: 0.217), suggesting limitations in handling dual bioactivity objectives. MAMO’s superior performance on both properties highlights its ability to coordinate expert agents for biological optimization tasks, particularly when the targets are interdependent. In the non-biological task (QED + SA), multiple methods achieve competitive results. MAMO attains the best average QED score (0.806) and the second-best SA score (0.813), ranking second overall. REINVENT Trans slightly outperforms MAMO in average SA (0.854), but lags in QED. These results indicate that while traditional models may excel in optimizing individual non-biological objectives, MAMO maintains strong and balanced performance across both, demonstrating its capability for general-purpose multi-objective optimization. Taken together, these findings confirm that MAMO effectively balances trade-offs across diverse molecular properties, particularly in challenging biological settings where traditional or single-agent methods struggle to generalize.

Pareto Front Analysis in Objective Space. To visually assess the optimization performance, we plot the non-dominated solution sets obtained by different methods on three representative tasks: $\text{GSK3}\beta$ + JNK3, QED + SA, and $\text{GSK3}\beta$ + QED. As shown in Figure 4, MAMO consistently outperforms baselines in both convergence and di-

versity across the objective space. Compared to other methods, MAMO’s solutions exhibit broader coverage and a more uniform distribution along the Pareto front, indicating stronger trade-off management between conflicting objectives. In particular, the improvement is most pronounced on biologically relevant tasks such as $\text{GSK3}\beta$ + JNK3 and $\text{GSK3}\beta$ + QED, where the optimization landscape is more challenging due to intricate interdependencies. In contrast, baseline methods often concentrate in limited regions or suffer from sparsity, failing to approximate the full Pareto front. These results further demonstrate the effectiveness of MAMO’s multi-agent coordination mechanism, which adaptively adjusts optimization strategies in response to dynamic conflicts across objectives. Overall, the experimental results demonstrate the strong advantages of MAMO in tackling high-dimensional and highly conflicting optimization tasks. In particular, its dynamic multi-agent collaboration mechanism proves crucial in scenarios where target objectives exhibit strong interdependencies.

Ablation Study. To investigate the importance of the central scheduling agent in MAMO, we designed ablation experiments by replacing its dynamic coordination with predefined sequential optimization chains and by removing inter-agent collaboration. Specifically, we evaluated MAMO on two tasks of increasing complexity: the three-objective task ($\text{GSK3}\beta$ +JNK3+QED) and the four-objective task ($\text{GSK3}\beta$ +JNK3+QED+SA). In the sequential setting, agents optimize objectives one at a time in a fixed order without global context or real-time feedback.

Methods	Top-10 Avg QED	Top-10 Avg SA	Top-50 Avg QED	Top-50 Avg SA	Avg QED	Avg SA	Avg Rank
REINVENT	0.944	0.936	0.930	0.920	0.725	0.790	5
Graph GA	0.943	0.996	0.930	0.968	0.685	0.791	8
SMILES GA	0.939	0.908	0.926	0.877	0.761	0.735	6
GP BO	0.939	0.973	0.926	0.954	0.702	<u>0.816</u>	4
REINVENT Trans	<u>0.944</u>	0.979	0.937	<u>0.962</u>	<u>0.790</u>	0.854	1
DrugAssist	0.941	0.969	0.927	0.931	0.731	0.789	3
EAG	0.931	0.932	0.925	0.939	0.712	0.767	7
GPT-4o mini	0.877	0.916	0.833	0.894	0.576	0.793	9
MAMO	0.945	<u>0.985</u>	<u>0.933</u>	0.968	0.798	0.813	2

Table 4: Performance of MAMO on top-ranked molecules under QED+SA objectives.

Optimization Chain	GSK3 β +JNK3+QED	GSK3 β +JNK3+QED+SA
GSK3 β -JNK3-QED	0.760	-
QED-JNK3-GSK3 β	0.572	-
GSK3 β -JNK3-QED-SA	-	0.501
QED-JNK3-GSK3 β -SA	-	0.510
MAMO_NoCo	0.580	0.530
MAMO	0.762	0.671

Table 5: HV of MAMO under different optimization settings. ‘-’ indicates configurations that are not applicable.

In the no-collaboration setting, each agent optimizes the SMILES independently without sharing optimization states across agents. As shown in Table 5, the complete MAMO framework consistently outperforms all sequential variants and the no-collaboration setting (MAMO_NOCO), achieving the highest HV scores on both tasks. The performance gap is particularly pronounced in the four-objective setting, where the absence of dynamic scheduling or agent coordination results in substantial declines. For example, the HV drops from 0.671 (MAMO) to 0.501/0.510 in sequential baselines and 0.530 in MAMO_NOCO, revealing the inefficiency of fixed update orders and isolated optimization in high-conflict scenarios. These results demonstrate that both sequential chains and independent optimization without collaboration cannot effectively coordinate multiple agents, often leading to conflicting or redundant updates. The lack of interaction across objectives and intermediate state sharing limits global optimization, especially when objectives are interdependent. In contrast, the MAMO central scheduling agent enables agents to exchange intermediate outputs and dynamically adjust their optimization focus based on task needs. This flexible orchestration promotes cooperative behavior and improves convergence toward the Pareto front.

Interpretable Optimization Path

We illustrate the iterative optimization process of the MAMO framework on a pyridine-based molecule (initial SMILES: Cc1ccn(Nc2cc(C)nc(-c3ccccc3)n2)c1). As shown in Figure 5, MAMO performs multi-round optimization to progressively improve molecular attributes. In the first iteration, the left-side pyridine ring is modified by replacing the methyl substituent with a hydrogen atom, and the central pyrazole loses its methyl group. Simultaneously, the right phenyl ring is substituted with a pyridine ring. These changes jointly lead to a strong boost in biological activ-

ity scores (GSK3 β and JNK3) and drug-likeness (QED). In the second iteration, the right pyridine is replaced with a phenyl group, further enhancing JNK3 and synthetic accessibility (SA), with only a minor QED decrease. In the final iteration, the right ring is modified again, and the remaining methyl group on the central scaffold is removed. This results in a balanced gain across all properties, notably in GSK3 β , JNK3, and SA. This trajectory showcases how MAMO adaptively coordinates expert agents to perform structure edits, achieving a consistent upward trend in overall molecular quality through goal-aware collaboration.

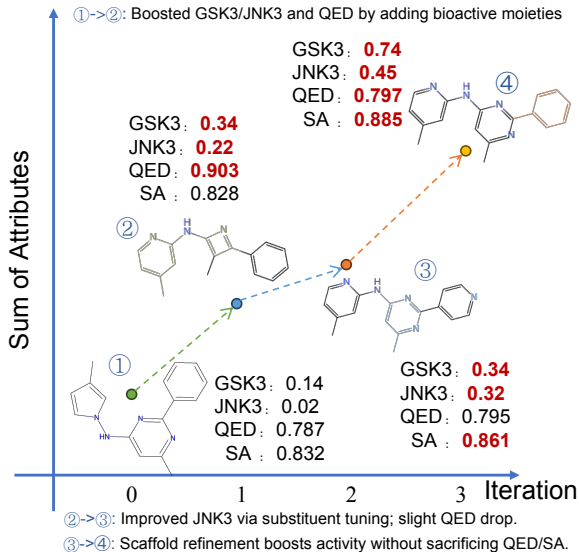


Figure 5: Molecular Optimization Trajectory with MAMO.

Conclusion

We propose MAMO, a novel multi-agent framework for multi-objective molecular optimization that resolves inter-objective conflicts through a centralized scheduling agent. By dynamically coordinating agent collaboration, MAMO achieves efficient optimization even in high-dimensional, conflicting objective spaces. Extensive experiments validate its effectiveness and robustness. Future work will focus on improving its scalability and extending its applicability to broader molecular design challenges.

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Technical Appendix

A Expert Prompt Design

This section provides a detailed introduction to the prompt design templates of each expert agent and the tool components used.

When constructing prompts, we follow the example formats in the original open-source code as closely as possible to ensure that the LLM can correctly interpret and execute the tasks. To ensure stable system operation, we strictly standardize the output format of each expert agent. Fig. 1, 2, and 3, respectively, present the prompt templates of the GSK3 β expert, the JNK3 expert, and the SA expert. Notably, for the SA (Synthetic Accessibility) property, we apply normalization to scale its range to [0, 1], and convert the minimization problem into a maximization one. We explicitly state this in the prompt to avoid misunderstandings caused by discrepancies between the pre-training knowledge of the model and the task setup. Specifically, the closer the SA value is to 1, the easier the molecule is to synthesize; the closer it is to 0, the more difficult the synthesis.

The MAMO framework primarily integrates two key tools: RDKit and Oracle.

RDKit is a powerful open-source cheminformatics toolkit used for the processing, analysis, and visualization of molecular data. It supports various molecular representation formats (such as SMILES, SMARTS, and SDF), and is widely used in molecular construction, manipulation, and optimization tasks.

The Oracle library is an integrated set of molecular evaluation tools widely used for molecular property prediction and drug discovery workflows. It supports molecular property computation and similarity evaluation, enabling the rapid screening of candidate compounds similar to the target structure based on molecular fingerprints or structural features, which is of significant value in virtual screening and drug design.

B More Details On Strategy Formulation

As shown in Fig. 4, we present the prompt for the Strategy Planner.

C More Details On Evaluation

As shown in Fig. 5, we present the prompt for the evaluation agent.

In the evaluation agent, tool utilization plays a crucial role. Because the agent cannot directly access the specific properties of the candidate molecules, it relies on tools to make predictions. We primarily use two tools: Oracle and RDKit.

Oracle is commonly used in the field of molecular property prediction as a property evaluator. Its primary function is to assess the performance of molecules by taking molecular structures as input and submitting their corresponding property values. Specifically, Oracle can evaluate a wide range of molecular objectives.

RDKit is an open source path-formatics toolkit designed for researchers in pathformatics and drug discovery. Pro-



GSK3 β Expert Prompt

You are a molecular optimization expert specializing in designing molecules with enhanced binding affinity to GSK3 β (Glycogen Synthase Kinase 3 Beta). GSK3 β binding affinity is typically measured through IC50 values, with lower values indicating stronger binding. The goal is to optimize the molecular structure to reduce the IC50 value while maintaining an acceptable balance of other pharmaceutical properties such as synthetic accessibility, drug-likeness, and toxicity.

Task Description:

Based on the provided molecular information and optimization strategies, use a Chain-of-Thought (COT) strategy generation mechanism to systematically optimize the molecular structure in stages to maximize its binding affinity to GSK3 β . Ensure the optimization enhances the binding interactions (e.g., through hydrophobic interactions, hydrogen bonding, or π -stacking) while balancing other critical pharmaceutical properties. The resulting molecule should be feasible for drug development.

Optimization Strategies:

{Generated through the optimization strategy mechanism}

Input Information:

SMILES: {}

Output Requirements:

Provide the optimized molecule SMILES as a list, returning only the SMILES strings in the following format:

['smiles1', 'smiles2', 'smiles3', ...]

The list should only contain SMILES strings without any additional content or explanation.

Figure 1: Prompt Design Template for the GSK3 β Expert Agent.



JNK3 Expert Prompt

You are a molecular optimization expert specializing in designing molecules with enhanced binding affinity to GSK3 β (Glycogen Synthase Kinase 3 Beta). GSK3 β binding affinity is typically measured through IC50 values, with lower values indicating stronger binding. The goal is to optimize the molecular structure to reduce the IC50 value while maintaining an acceptable balance of other pharmaceutical properties such as synthetic accessibility, drug-likeness, and toxicity.

Task Description:

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Optimization Strategies:

{Generated through the optimization strategy mechanism}

Input Information:

SMILES: {}

Output Requirements:

Provide the optimized molecule SMILES as a list, returning only the SMILES strings in the following format:

['smiles1', 'smiles2', 'smiles3', ...]

The list should only contain SMILES strings without any additional content or explanation.

Figure 2: Prompt Design Template for the JNK3 Expert Agent.

vides a comprehensive software framework for processing, analyzing, and visualizing molecular data.

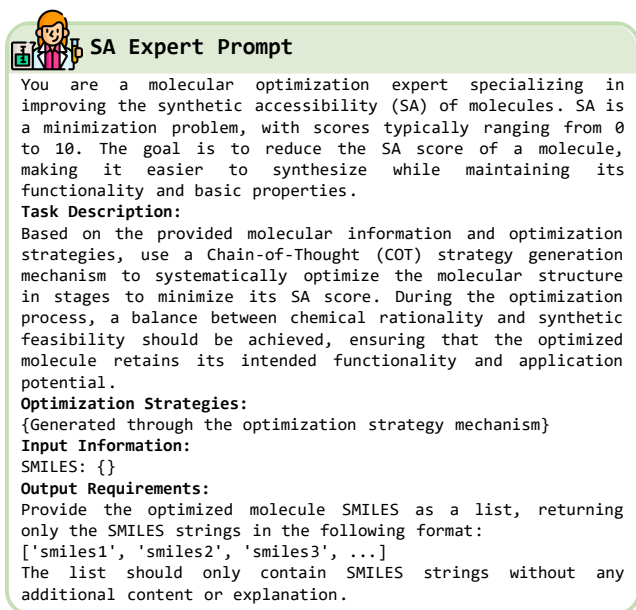


Figure 3: Prompt Design Template for the SA Expert Agent.

D More Details On Agent Scheduling

Prompt As shown in Fig. 6, we present the prompt for the central scheduling agent.

Central Scheduling with Binary Integer Programming

The Binary Integer Programming (BIP) formulation in Equation 6 operationalizes our multi-agent collaboration framework through dynamic task allocation. Its objective function maximizes the sum of priority scores:

$$\max_{A \in \{0,1\}^{N \times K}} \sum_{i=1}^N \sum_{j=1}^K A_{i,j} \pi_i^t \quad (1)$$

where π_i^t represents the normalized optimization gap of molecule m_i (Equation 5), ensuring optimal alignment between molecules and expert agents based on real-time evaluation feedback.

The constraints implement collaborative resource allocation:

- $\sum_{i=1}^N A_{i,j} = 1$ ensures each agent receives exactly one molecule per round, preventing agent idling
- $\sum_{j=1}^K A_{i,j} \leq 1$ restricts molecules to at most one agent, avoiding redundant optimizations

We implement this scheduling mechanism using Python’s PuLP library with default solvers. The BIP solution typically converges within 0.2 seconds per iteration - enabling responsive coordination essential for our collaborative optimization paradigm. This efficient dynamic reassignment allows expert agents to continuously refocus their efforts on the most critical molecular improvements.

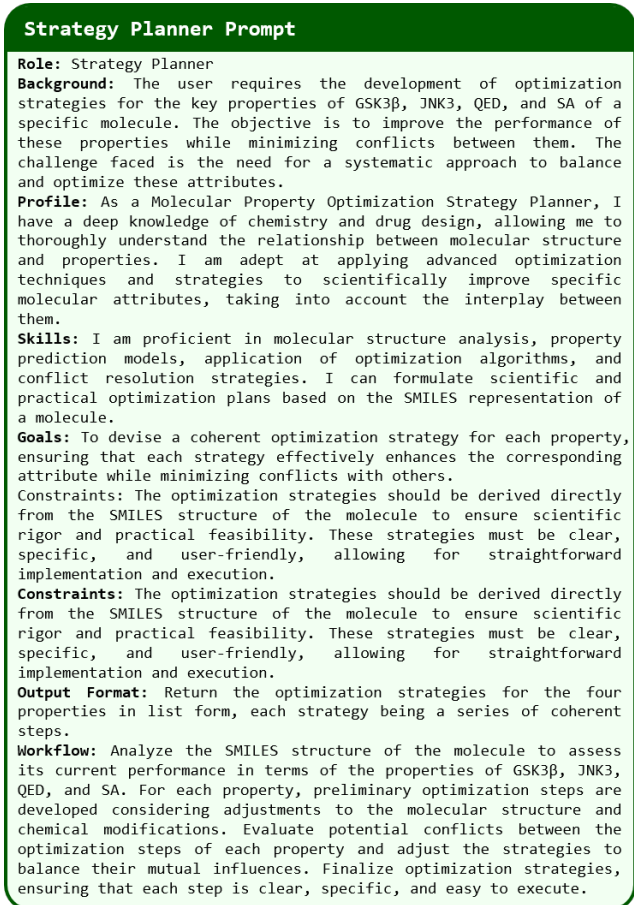


Figure 4: Prompt Design Template for the Strategy Planner.

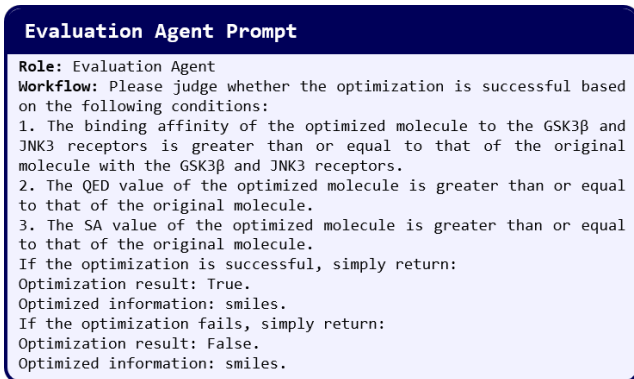


Figure 5: Prompt Design Template for the Evaluation Agent.

E Introduction To Related Properties

GSK3β. Glycogen synthase kinase 3 beta (GSK3β) is a serine/threonine protein kinase involved in various cellular processes, including glycogen metabolism, cell proliferation, differentiation, and apoptosis. It has gained significant attention in neurodegenerative disease research due to its role in regulating tau protein phosphorylation, amyloid precursor

Table 1: Comparison of Div and Nov for Different Methods in Three-Objective and Four-Objective Scenarios.

Method	GSK3 β +JNK3+QED+SA		GSK3 β +JNK3+QED		GSK3 β +QED+SA		JNK3+QED+SA	
	Nov	Div	Nov	Div	Nov	Div	Nov	Div
GP BO	0.958	0.828	0.865	0.873	0.864	0.878	0.865	0.884
Graph GA	0.998	0.816	0.960	0.871	0.960	0.876	0.958	0.869
REINVENT	0.998	0.857	0.997	0.872	0.997	0.863	0.995	0.858
REINVENT Trans	0.971	0.873	0.998	0.845	0.994	0.822	0.996	0.810
SMILES GA	0.903	0.880	0.960	0.844	0.913	0.845	0.943	0.854
GPT-4o mini	0.997	0.867	0.997	0.861	0.997	0.869	0.997	0.866
MAMO	0.999	0.874	0.999	0.874	0.998	0.877	0.998	0.871

Central Scheduling Agent

As a molecular optimization decision expert, your role is to reassign optimization tasks based on the molecules currently optimized by the GSK3 β , JNK3, QED, and SA agents. The goal is to further enhance the overall performance of the molecules.

Task Description: Reassign the molecules for the next round of optimization based on the following information:

1. GSK3 β Agent: Optimize the molecule to increase binding affinity with GSK3 β .
2. JNK3 Agent: Optimize the molecule to increase binding affinity with JNK3.
3. QED Agent: Optimize the molecule to enhance its drug development potential (QED).
4. SA Agent: Optimize the molecule to increase the normalized SA value (converted to a maximization problem within the [0, 1] range, where values closer to 1 indicate easier synthesis).

Current Molecule Information:

Molecule optimized by the GSK3 β Expert Agent: {}
Molecule optimized by the JNK3 Expert Agent: {}
Molecule optimized by the QED Expert Agent: {}
Molecule optimized by the SA Expert Agent: {}

Allocation Principles:

1. Assign molecules optimized by each agent to other agents that have not yet optimized them. This ensures that each molecule undergoes comprehensive optimization from different perspectives.
2. Prevent assigning the same molecule to a single agent for two consecutive rounds to maintain diversity and the global nature of optimization tasks.
3. Ensure that the allocation aligns with the molecular optimization direction and the goals of each agent for multi-dimensional improvement of the molecules.
4. Avoid assigning the same molecule to the same agent in consecutive rounds.

Output Format:

Next Round of Molecular Optimization Task Allocation:
GSK3 β Agent: smilesX
JNK3 Agent: smilesY
QED Agent: smilesZ
SA Agent: smilesW

Figure 6: Prompt Design Template for the central scheduling Agent.

protein processing, and neuronal survival. Inhibiting GSK3 β is considered a vital molecular objective in drug discovery, as it could modulate these pathological processes and potentially slow or prevent the progression of neurodegenerative diseases.

JNK3. JNK3 is a member of the c-Jun N-terminal kinases (JNKs) family, which belongs to the mitogen-activated protein kinase (MAPK) pathway and is primarily expressed in the central nervous system. It plays a crucial role in mediating cellular responses to stress, including apoptosis, in-

flammation, and neuronal damage. Targeting JNK3 inhibition is one of the key molecular objectives in drug discovery because it may prevent or reduce neuronal cell death and inflammation, making it a promising therapeutic target for neurodegenerative diseases such as Alzheimer’s disease.

QED. Quantitative Estimate of Drug-likeness (QED) is a metric widely used in molecular optimization to evaluate the drug-likeness of a molecule. It consists of several physico-chemical properties, including molecular weight, lipophilicity (logP), topological polar surface area (TPSA), the number of hydrogen bond donors and acceptors, and the count of aromatic rings and rotatable bonds. It provides a score ranging from 0 to 1, with higher scores indicating molecules that are more likely to have favorable drug-like properties.

SA. Synthetic Accessibility (SA) is a metric used in molecular optimization to assess the ease with which a molecule can be synthesized in a laboratory setting. It considers various structural features that influence synthesis complexity, such as the presence of complex ring systems, functional groups, stereocenters, and the overall size and branching of the molecule. The SA score ranges from 1 to 10, with lower scores indicating higher synthetic feasibility. In this work, we transform the SA score into the normalized SA score, following prior studies, to formulate it as a maximization objective. Specifically, the normalized SA score is given by the following equation:

$$Normalized_SA = 1 - \frac{SA - 1}{9} \quad (2)$$

This adjustment ensures that higher normalized SA scores correspond to molecules that are easier to synthesize, within the score range of 0 to 1.

F More Details On Dataset and Metrics

Dataset In our experiments, we initialize the population with 100 molecules randomly sampled from the Ratio-naleRL dataset. For each optimization cycle, the top-15 scoring molecules generated by MAMO are selected as candidate solutions.

Comparison Using Traditional Evaluation Metrics Novelty (Nov) quantifies the percentage of molecules generated that are dissimilar (similarity less than 0.4) to the nearest neighbor in a reference molecule library. This metric assesses the uniqueness of the generated molecules.

Diversity (Div) quantifies the dissimilarity between pairs of generated molecules. It is calculated using pairwise Tanimoto similarity over Morgan fingerprints. Div reflects the structural diversity of the generated molecules.

Results. Table 5 presents the average Nov and Div values in five runs for various methods in optimization tasks with three objectives and four objectives. These results highlight several key performance advantages of MAMO over baseline methods: 1) In all tasks, MAMO consistently achieved Nov values higher than those of all baseline methods, demonstrating its exceptional ability to generate molecules with high structural novelty. This indicates that MAMO is capable of producing a greater number of novel molecules that significantly differ from the reference molecular library. 2) MAMO also outperformed most baseline models in Div values across all tasks, especially excelling in the GSK3 β +JNK3+QED and GSK3 β +QED+SA tasks. This demonstrates MAMO’s capability to maintain structural diversity while optimizing multi-objective performance. 3) MAMO not only excelled in both Nov and Div metrics but also displayed remarkable stability and consistency across all tasks. This highlights its ability to provide a set of molecules that perform exceptionally well in terms of effectiveness, diversity, and novelty in multi-objective optimization. In contrast, baseline methods exhibited significant performance fluctuations in certain tasks, making MAMO’s advantages even more comprehensive and prominent.

G More Details Of the experimental setup of GPT-4o mini.

GPT-4o mini Prompt

Role: Molecular optimization expert.
Task Description:
 According to the following optimization objectives, optimize the given molecule. The optimized molecule should be better than the pre-optimized molecule in terms of the properties to be optimized, and ensure that the generated candidate molecule does not duplicate the previously generated one.
Optimization Objective:
 GSK3 β , JNK3, QED and SA.
Previously optimized information:
 {Generated through the optimization strategy mechanism}
 Number of iterations: 1. Generate candidate molecules: ['smiles1', 'smiles2', 'smiles3', ...]
 Number of iterations: 2. Generate candidate molecules: ['smiles1', 'smiles2', 'smiles3', ...]
 Number of iterations: 3...
Input Information:
 SMILES: {}
Output Requirements:
 Provide the optimized molecule SMILES as a list, returning only the SMILES strings in the following format:
 ['smiles1', 'smiles2', 'smiles3', ...]
 The list should only contain SMILES strings without any additional content or explanation.

Figure 7: Prompt Design Template for the Gpt-4o mini.

In this study, we investigate the impact of the number of iterations on the performance of a multi-agent collaborative system in a four-objective optimization task (GSK3B, JNK3, QED, and SA) and compare MAMO with GPT-4o mini. As shown in Fig. 7, we present the prompt design template for GPT-4o mini. In the experimental setup, GPT-4o mini

strictly avoids generating duplicate molecules in each iteration and incorporates the optimization results of each round into its memory to facilitate the accumulation and sharing of information.

H More Details Of Baseline Methods

We conducted experiments using the following baseline methods: GP BO, Graph GA, REINVENT, REINVENT Trans, and SMILES GA. The specific experimental setup and details are as follows:

1) Initial Population Setup To ensure fairness and comparability of the results, we use an initial population consistent with that of MAMO in all baseline method experiments. This consistency ensures that the performance comparison across methods is not influenced by initial conditions, allowing for a more objective assessment of each method’s optimization capability.

2) Parameter Configuration For GP BO, Graph GA, REINVENT, and SMILES GA, we directly use the default parameter configurations provided in the PMO benchmark. These parameters have been validated by the community to ensure stable performance of the baseline methods in standard tasks. For REINVENT Trans, due to the absence of default parameter configurations, we opted to retrain the model for a fair comparison. The specific training details are shown in Table 6, and the experimental environment and the parameter settings are as follows.

Table 2: Training parameters of REINVENT Trans.

Hardware	NVIDIA 3080 with 24GB memory
Training Epochs	75
Time	37.5h
Learning Rate	0.0005
Batch Size	32
Sigma	500
Experience Replay	24

Through the aforementioned setup, we ensured that the baseline methods performed at their maximum potential in the experiments, providing a fair and reliable basis for comparison with the performance of MAMO.